

IdM 修复

- `student@workstation` 上执行 `lab start security-ldap` 报错本质是由于 IdM 证书过期导致的平台不可服务。
- 因此，登录 `root@utility` 修复 IdM:

```
systemctl status dirsrv@LAB.EXAMPLE.COM.service      #确保服务 active
ipa-cert-fix -v --log-file=/tmp/ipa_cert_fix.log      #输入 yes 继续

certutil -L -d /etc/pki/pki-tomcat/alias -a -n "LAB.EXAMPLE.COM IPA CA" -o
/tmp/ipa-ca.crt
mv /etc/ipa/ca.crt /etc/ipa/ca.crt.old
cp /tmp/ipa-ca.crt /etc/ipa/ca.crt
chmod 0644 /etc/ipa/ca.crt
cp /tmp/ipa-ca.crt /etc/pki/ca-trust/source/anchors/ipa-ca.crt
update-ca-trust extract

mv /var/lib/ipa-client/pki/kdc-ca-bundle.pem /var/lib/ipa-client/pki/kdc-ca-
bundle.pem.old
cp /tmp/ipa-ca.crt /var/lib/ipa-client/pki/kdc-ca-bundle.pem
chmod 0644 /var/lib/ipa-client/pki/kdc-ca-bundle.pem
systemctl restart httpd.service krb5kdc.service

ipactl status
kinit admin@LAB.EXAMPLE.COM      #密码: RedHat123^
ipa host-show utility.lab.example.com
#可返回主机信息条目表示成功
#同时登录 https://utility.lab.example.com, 用户名 admin, 密码 RedHat123^, 登录成功
表示修复完成。
```

```
[root@utility ~]# systemctl status dirsrv@LAB-EXAMPLE-COM.service
● dirsrv@LAB-EXAMPLE-COM.service - 389 Directory Server LAB-EXAMPLE-COM.
   Loaded: loaded (/usr/lib/systemd/system/dirsrv@.service; enabled; vendor preset: disabled)
   Drop-In: /usr/lib/systemd/system/dirsrv@.service.d
            └─custom.conf
            /etc/systemd/system/dirsrv@LAB-EXAMPLE-COM.service.d
            └─ipa-env.conf
   Active: active (running) since Thu 2026-06-18 04:25:08 EDT; 6min ago
   Process: 1292 ExecStartPre=/usr/libexec/dirsrv/ds_systemd_ask_password_acl /etc/dirsrv/slapd-LAB-EXAMPLE-COM/
Main PID: 1306 (ns-slapd)
   Status: "slapd started: Ready to process requests"
     Tasks: 28 (limit: 36237)
    Memory: 139.4M
    CGroup: /system.slice/system-dirsrv.slice/dirsrv@LAB-EXAMPLE-COM.service
            └─1306 /usr/sbin/ns-slapd -D /etc/dirsrv/slapd-LAB-EXAMPLE-COM -i /run/dirsrv/slapd-LAB-EXAMPLE-COM.

Jun 18 04:25:08 utility.lab.example.com ns-slapd[1306]: [18/Jun/2026:04:25:08.316903834 -0400] - ERR - cos-plug
Jun 18 04:25:08 utility.lab.example.com ns-slapd[1306]: [18/Jun/2026:04:25:08.349267628 -0400] - ERR - schema-c
Jun 18 04:25:08 utility.lab.example.com ns-slapd[1306]: [18/Jun/2026:04:25:08.359279976 -0400] - INFO - slapd_d
Jun 18 04:25:08 utility.lab.example.com ns-slapd[1306]: [18/Jun/2026:04:25:08.363329513 -0400] - INFO - slapd_d
Jun 18 04:25:08 utility.lab.example.com ns-slapd[1306]: [18/Jun/2026:04:25:08.367156233 -0400] - INFO - slapd_d
Jun 18 04:25:08 utility.lab.example.com systemd[1]: Started 389 Directory Server LAB-EXAMPLE-COM..
Jun 18 04:25:13 utility.lab.example.com ns-slapd[1306]: [18/Jun/2026:04:25:13.372375775 -0400] - ERR - schema-c
Jun 18 04:25:13 utility.lab.example.com ns-slapd[1306]: [18/Jun/2026:04:25:13.396255005 -0400] - ERR - schema-c
Jun 18 04:25:13 utility.lab.example.com ns-slapd[1306]: [18/Jun/2026:04:25:13.557871352 -0400] - ERR - schema-c
Jun 18 04:25:13 utility.lab.example.com ns-slapd[1306]: [18/Jun/2026:04:25:13.576710969 -0400] - ERR - schema-c
```

```

Becoming renewal master.
ipalib.sysrestore: DEBUG: Loading StateFile from '/var/lib/ipa/sysrestore/sysrestore.state'
ipalib.sysrestore: DEBUG: Loading Index file from '/var/lib/ipa/sysrestore/sysrestore.index'
Restarting IPA
ipapython.ipauttil: DEBUG: Starting external process
ipapython.ipauttil: DEBUG: args=['ipactl', 'restart']
ipapython.ipauttil: DEBUG: Process finished, return code=0
ipapython.ipauttil: DEBUG: stdout=Restarting Directory Service
Restarting krb5kdc Service
Restarting kadmin Service
Restarting httpd Service
Restarting ipa-custodia Service
Restarting pki-tomcatd Service
Restarting ipa-otpd Service

ipapython.ipauttil: DEBUG: stderr=ipa: INFO: The ipactl command was successful
ipapython.admintool: INFO: The ipa-cert-fix command was successful

```

```

[root@utility ~]# certutil -L -d /etc/pki/pki-tomcat/alias -a -n "LAB.EXAMPLE.COM IPA CA" -o /tmp/ipa-ca.crt
[root@utility ~]# mv /etc/ipa/ca.crt /etc/ipa/ca.crt.old
[root@utility ~]# cp /tmp/ipa-ca.crt /etc/ipa/ca.crt
[root@utility ~]# chmod 0644 /etc/ipa/ca.crt
[root@utility ~]# cp /tmp/ipa-ca.crt /etc/pki/ca-trust/source/anchors/ipa-ca.crt
[root@utility ~]# update-ca-trust extract
[root@utility ~]#
[root@utility ~]# mv /var/lib/ipa-client/pki/kdc-ca-bundle.pem /var/lib/ipa-client/pki/kdc-ca-bundle.pem.old
[root@utility ~]# cp /tmp/ipa-ca.crt /var/lib/ipa-client/pki/kdc-ca-bundle.pem
[root@utility ~]# chmod 0644 /var/lib/ipa-client/pki/kdc-ca-bundle.pem
[root@utility ~]# systemctl restart httpd.service krb5kdc.service
[root@utility ~]#
[root@utility ~]# ipactl status
Directory Service: RUNNING
krb5kdc Service: RUNNING
kadmin Service: RUNNING
httpd Service: RUNNING
ipa-custodia Service: RUNNING
pki-tomcatd Service: RUNNING
ipa-otpd Service: RUNNING
ipa: INFO: The ipactl command was successful

```

```

[root@utility ~]# kinit admin@LAB.EXAMPLE.COM
Password for admin@LAB.EXAMPLE.COM:
[root@utility ~]# ipa host-show utility.lab.example.com
Host name: utility.lab.example.com
Principal name: host/utility.lab.example.com@LAB.EXAMPLE.COM
Principal alias: host/utility.lab.example.com@LAB.EXAMPLE.COM
SSH public key fingerprint: SHA256:GDY6Hz6wbF6LuddHSLMjqEc9m+BiDPTsoYouJcJl0CY (ssh-ed25519),
SHA256:NJAyJMx8B2AeIYHRnVLAuJlXZwblomyOKowyfTwGrTY (ecdsa-sha2-nistp256),
SHA256:k/ZgwRwPCOp6YbGHRpYBN3Z2IME25FkRDNoH50V6vps (ssh-rsa)

Password: False
Member of host-groups: ipaservers
Keytab: True
Managed by: utility.lab.example.com
[root@utility ~]#

```

- 登录 `student@workstation` 执行 `lab start security-ldap` 即可成功。
- 登录 `root@servera` 修改 `sssd` 相关配置文件权限，否则依然无法修复 `sssd` 本身权限问题：

```

chmod 0755 /etc/sssd/
chmod 0600 /etc/sssd/sssd.conf
scp root@utility:/etc/krb5.keytab /etc/krb5.keytab
#如果不同步此文件，即使按照教材练习完成步骤，依然 sssd 服务重启失败，报错如下图。

```

```
Jun 18 04:57:41 servera.lab.example.com sssd[be[lab.example.com]][26592]: krb5_kt_start_seq_get failed: Key tabl
e file '/etc/krb5.keytab' not found
Jun 18 04:57:41 servera.lab.example.com sssd[be[lab.example.com]][26592]: krb5_kt_start_seq_get failed: Key tabl
e file '/etc/krb5.keytab' not found
Jun 18 04:57:41 servera.lab.example.com sssd[be[lab.example.com]][26592]: krb5_kt_start_seq_get failed: Key tabl
e file '/etc/krb5.keytab' not found
Jun 18 04:57:41 servera.lab.example.com sssd[be[lab.example.com]][26592]: krb5_kt_start_seq_get failed: Key tabl
e file '/etc/krb5.keytab' not found
Jun 18 04:57:41 servera.lab.example.com sssd[be[lab.example.com]][26592]: krb5_kt_start_seq_get failed: Key tabl
e file '/etc/krb5.keytab' not found
Jun 18 04:57:41 servera.lab.example.com sssd[be[lab.example.com]][26592]: krb5_kt_start_seq_get failed: Key tabl
e file '/etc/krb5.keytab' not found
Jun 18 04:57:41 servera.lab.example.com sssd[be[lab.example.com]][26592]: krb5_kt_start_seq_get failed: Key tabl
e file '/etc/krb5.keytab' not found
Jun 18 04:57:41 servera.lab.example.com sssd[be[lab.example.com]][26592]: krb5_kt_start_seq_get failed: Key tabl
e file '/etc/krb5.keytab' not found
Jun 18 04:57:41 servera.lab.example.com sssd[be[lab.example.com]][26592]: Failed to read keytab [FILE:/etc/krb5.
keytab]: No suitable principal found in keytab
Jun 18 04:57:41 servera.lab.example.com sssd[be[lab.example.com]][26593]: Starting up
Jun 18 04:57:41 servera.lab.example.com sssd[be[lab.example.com]][26593]: krb5_kt_start_seq_get failed: Key tabl
e file '/etc/krb5.keytab' not found
```

- 按照教材放方法继续排错完成练习即可。